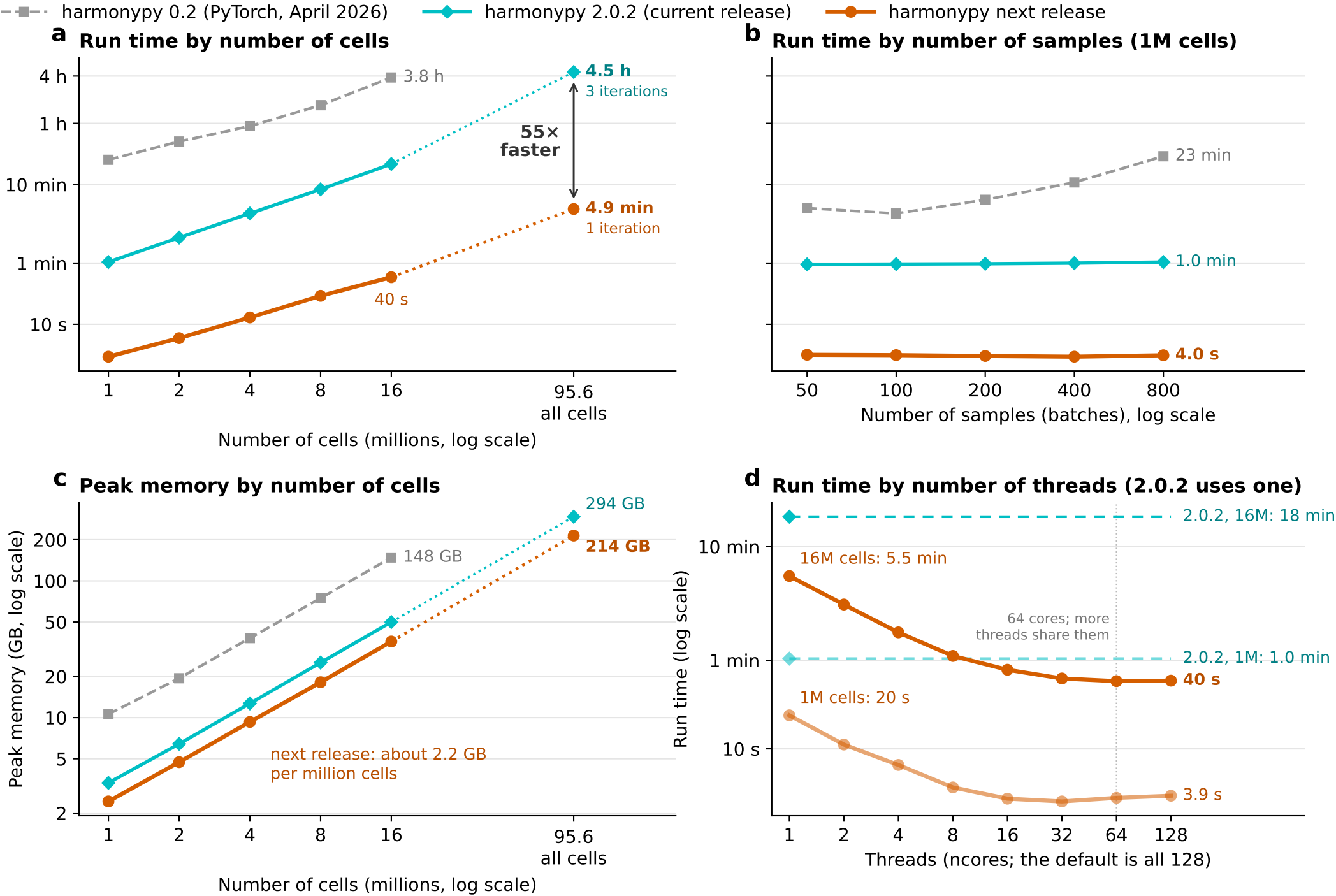


harmony next release on Tahoe-100M: all 95.6 million cells in 4.9 min

55× faster than 2.0.2 on all cells (4.5 h, where 2.0.2 needed 3 iterations and the next release 1);
28× faster on 16 million cells, with the same number of iterations; 3.3× faster even on one thread (16M cells);
same results as 2.0.2 (per-PC correlation ≥ 0.9999)



Data: Tahoe-100M, 50 PCs, corrected for sample; the 1–16 million cell subsamples have 800 samples, all 95.6 million cells have 1,344 (dotted segments). Default settings; for 2.0.2 and the next release, every run converged after 1 Harmony iteration except 2.0.2 on all cells (3). 0.2: April 2026 benchmark on the same machine, not run on all cells; its defaults do more clustering work (epsilon_harmony 1e-4 instead of 1e-2, up to 20 k-means rounds instead of 4), and its memory is what the process held after the run, not its peak. Run time: run_harmony plus reading the corrected coordinates (h.Z_corr), loading excluded (next release: median of 3–5 runs; 2.0.2: one to 3 runs, median). Peak memory: the whole process, input included. Machine: a shared server, 2× AMD EPYC 7543 (64 cores, 128 threads), 3 TB RAM. The next release (commit 623ff51) used all 128 threads, its default, except in d.